Comma categories

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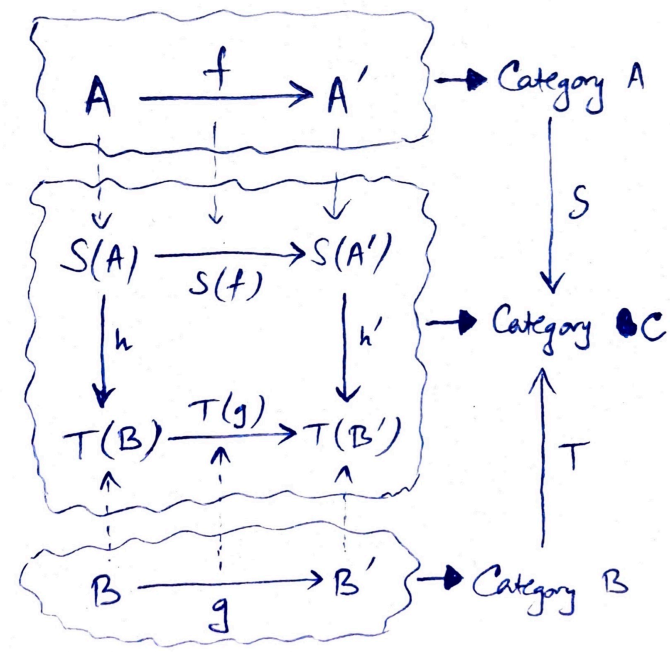
Have a look at the above page ***Free constructions*** *for a nice discussion.*

A comma category might be thought of as a way of talking about functions between objects which belong to 2 different categories, by bringing them on the same playing field, a common category. We will be trying to match the notation on this Wikipedia page: https://en.wikipedia.org/wiki/Comma_category

Suppose we have two categories $\boldsymbol{A}$ and $\boldsymbol{B}$, and we want to talk about functions of type $A \to B$, with $A \in \mathrm{obj}(\boldsymbol{A}), B \in \mathrm{obj}(\boldsymbol{B})$. We want to create a new category where the *objects* will be such functions.

First, we need a common category where can put the objects from $\boldsymbol{A}$ and $\boldsymbol{B}$ into. Suppose we have such a category $\boldsymbol{C}$ and 2 functors, $S : \boldsymbol{A} \to \boldsymbol{C}$ and $T : \boldsymbol{B} \to \boldsymbol{C}$. (They stand for "source" and "target", since S embeds the domains of the functions we want to define into $\boldsymbol{C}$, while T embeds the codomains).

Now, we define our category $S \downarrow T$. Its objects are tuples (A, B, h) with $A \in \mathrm{obj}(\boldsymbol{A}), B \in \mathrm{obj}(\boldsymbol{B}), h \in \mathrm{Hom}_{\boldsymbol{C}}(S(A), T(B))$. [Therefore the objects encode the function which we want along with information about its domain and codomain]. Morphisms between these "functions" are pairs of morphisms between their "domains" and "codomains" (such that they play compatibly with the functions); therefore, a morphism between (A, B, h) and (A', B', h') is a pair (f, g) with $f : A \to A', g : B \to B'$ such that $T(g) \circ h = h' \circ S(f)$.



$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[d, dashed, "S"] & A' \arrow[d, dashed, "S"] \\
S(A) \arrow[r, "S(f)"'] \arrow[d, "h"] & S(A') \arrow[d, "h'"] \\
T(B) \arrow[r, "T(g)"] & T(B') \\
B \arrow[r, "g"'] \arrow[u, dashed, "T"] & B' \arrow[u, dashed, "T"]
\end{tikzcd}$$

Fun fact!

The category of graphs $\mathbf{Gph}$ can be described as $(\mathrm{id}_{\mathbf{Set}} \downarrow D)$, where D is an endofunctor on $\mathbf{Set}$, mapping $S \mapsto S \times S$. Objects in $\mathbf{Gph}$ are triplets $(a, b, f : a \to b \times b)$, where a is the "edge indexing set", picking out its endpoints from the vertex set b.